CE2107 Microprocessor System Design and Development

Tutorial 2

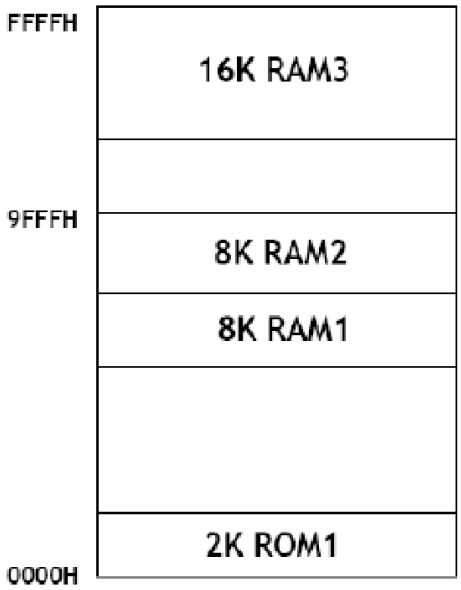
Signals and Interfacing

1. a) Explain how the bit banding is performed and show why 32MB of the address space is needed for the mapping of 1MB of bits.

   b) Show the codes that can be used to set b[3] (i.e. 4th bit) of address location 0x20000001.

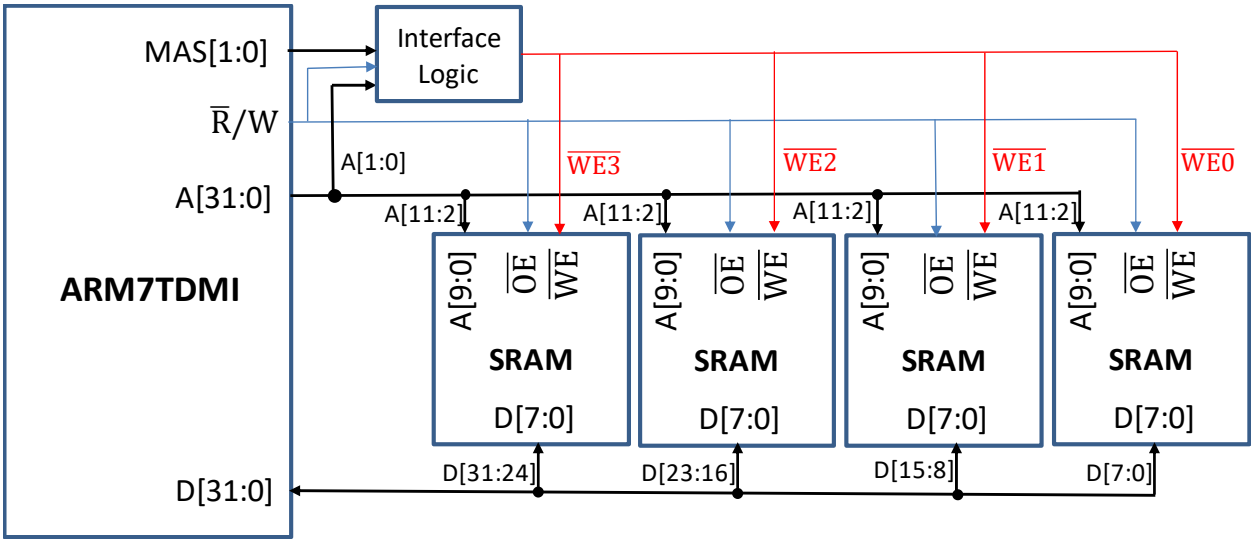
2. A microprocessor with 16-bit address bus needs to be interfaced with two 1Kx8 SRAMs.
   a) With the help of a diagram, show how partial address decoding can used for interfacing the microprocessor with the two SRAMs.
   b) Discuss the advantages and disadvantages of your address decoding design.

3. A microprocessor system has the memory map as shown below.
   a. Provide the starting and ending addresses for each memory device in hexadecimal format.
   b. Give the address decoding logic.
   c. Draw the address decoding circuit and provide hardware connections for the external memory interfacing.
      (Hint: Consider using a 3-to-8 decoder together with discrete logic gates)



4. This question is based on the memory interfacing diagram that is covered in the lecture notes. The microprocessor is interfaced to a 4Kx8 memory that is constructed using four 1Kx8 SRAMs.

   Design the interface logic that will allow you to **write** to the correct SRAM chips. MAS[1:0] can be used to determine if a byte/half-word/word is to be written to the SRAM based on the table below.



Access type	MAS[1:0]	A[1:0]	Little-endian
Word	10	XX	**D[31:0]**
Halfword	01	0X	**D[15:0]**
	01	1X	**D[31:16]**
Byte	00	00	**D[7:0]**
	00	01	**D[15:8]**
	00	10	**D[23:16]**
	00	11	**D[31:24]**

Optional

5. The following diagram shows the memory interface of a SRAM and an EPROM to a microprocessor that uses a memory selection signal PSEN# (Program Strobe Enable, active low) when accessing external memory. Specifically, PSEB# is asserted when the microprocessor is accessing the program code (rather than data).

   a. Describe the operation of the memory system with the microprocessor.
   b. A SRAM typically consumes higher power when its CE# is asserted (e.g., 100µA vs 2µA.) What is the disadvantage of this memory interface design and how it can be improved?

